THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The average of 38 numbers is 40. The average of all 57 numbers (including 19 more numbers) is 65. The average of the 19 additional numbers is: 1

(a) 116 **(b)** 114 **(c)** 117 **(d)** 115

2. A group of 28 people has average weight 68 kg. Another group has average weight 60 kg. Together the average weight is 62 kg. The number of people in the second group is: 1

(a) 85 **(b)** 83 **(c)** 86 **(d)** 84

3. 800 mL of a 12% sugar solution is mixed with 700 mL of a 57% sugar solution. The concentration of the mixture is: 1

(a) 33% **(b)** 35% **(c)** 34% **(d)** 32%

4. Marks of 97, 90 and 79 are given weights 7, 6 and 3 respectively. The weighted average is: 1

(a) 91 **(b)** 90 **(c)** 92 **(d)** 93

5. The average monthly salary of 20 employees is ₹59 thousand. An employee earning ₹33 thousand leaves and a new employee joins, and the average rises by ₹3 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 90 **(b)** 96 **(c)** 93 **(d)** 98

6. A shop earns a profit of 39% on sales of ₹7000 and 27% on sales of ₹5000. The overall profit percentage on total sales is: 1

(a) 35% **(b)** 36% **(c)** 34% **(d)** 33%

7. The average of 23 numbers is 75. When a number 51 is included, the average of all 24 numbers becomes: 1

(a) 73 **(b)** 74 **(c)** 75 **(d)** 76

8. The average of 19 numbers is 83. If the number 119 is removed, the average of the remaining 18 numbers is: 1

(a) 80 **(b)** 81 **(c)** 82 **(d)** 83

9. The average of 40 numbers is 65. The average of all 65 numbers (including 25 more numbers) is 70. The average of the 25 additional numbers is: 1

(a) 80 **(b)** 79 **(c)** 77 **(d)** 78

10. A group of 23 people has average weight 65 kg. Another group has average weight 85 kg. Together the average weight is 84 kg. The number of people in the second group is: 1

(a) 439 **(b)** 437 **(c)** 436 **(d)** 438

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The average monthly salary of 25 employees is ₹68 thousand. An employee earning ₹28 thousand leaves and a new employee joins; the average rises by ₹3 thousand. Find the new employee's monthly salary. 2

12. A cyclist rides from home to a market at 21 km/h and returns by the same route at 28 km/h. Find the average speed for the whole trip. 2

13. The mean of 40 observations is 62. One observation 101 is removed. Find the mean of the remaining observations. 2

14. The average monthly salary of 27 employees is ₹41 thousand. An employee earning ₹27 thousand leaves and a new employee joins; the average rises by ₹5 thousand. Find the new employee's monthly salary. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a class, 11 students have an average of 86 marks and the remaining students have an average of 74 marks. The class average is 77. Find the number of remaining students. 3

16. The average score of 34 students in a test is 63. When 51 more students take the test, the average score of all 85 students becomes 45. Find the average score of the 51 additional students. 3

17. 160 mL of a 42% acid solution is diluted by adding 80 mL of water. Find the new concentration. 3

OR

In a class, 17 students have an average of 35 marks and the remaining students have an average of 58 marks. The class average is 41. Find the number of remaining students. 3

18. The average score of 20 students in a test is 85. When 35 more students take the test, the average score of all 55 students becomes 64. Find the average score of the 35 additional students. 3

OR

70 mL of a 24% acid solution is diluted by adding 10 mL of water. Find the new concentration. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three sections have 37, 37 and 37 students with average marks 88, 86 and 60 respectively. Find the average marks of all the students. 5

OR

740 mL of a 43% sugar solution is mixed with 370 mL of a 31% sugar solution. Find the percentage concentration of the mixture. 5

20. Three sections have 21, 56 and 48 students with average marks 51, 45 and 58 respectively. Find the average marks of all the students. 5

OR

850 mL of a 22% sugar solution is mixed with 510 mL of a 54% sugar solution. Find the percentage concentration of the mixture. 5

END OF QUESTION PAPER — Total: 40 Marks